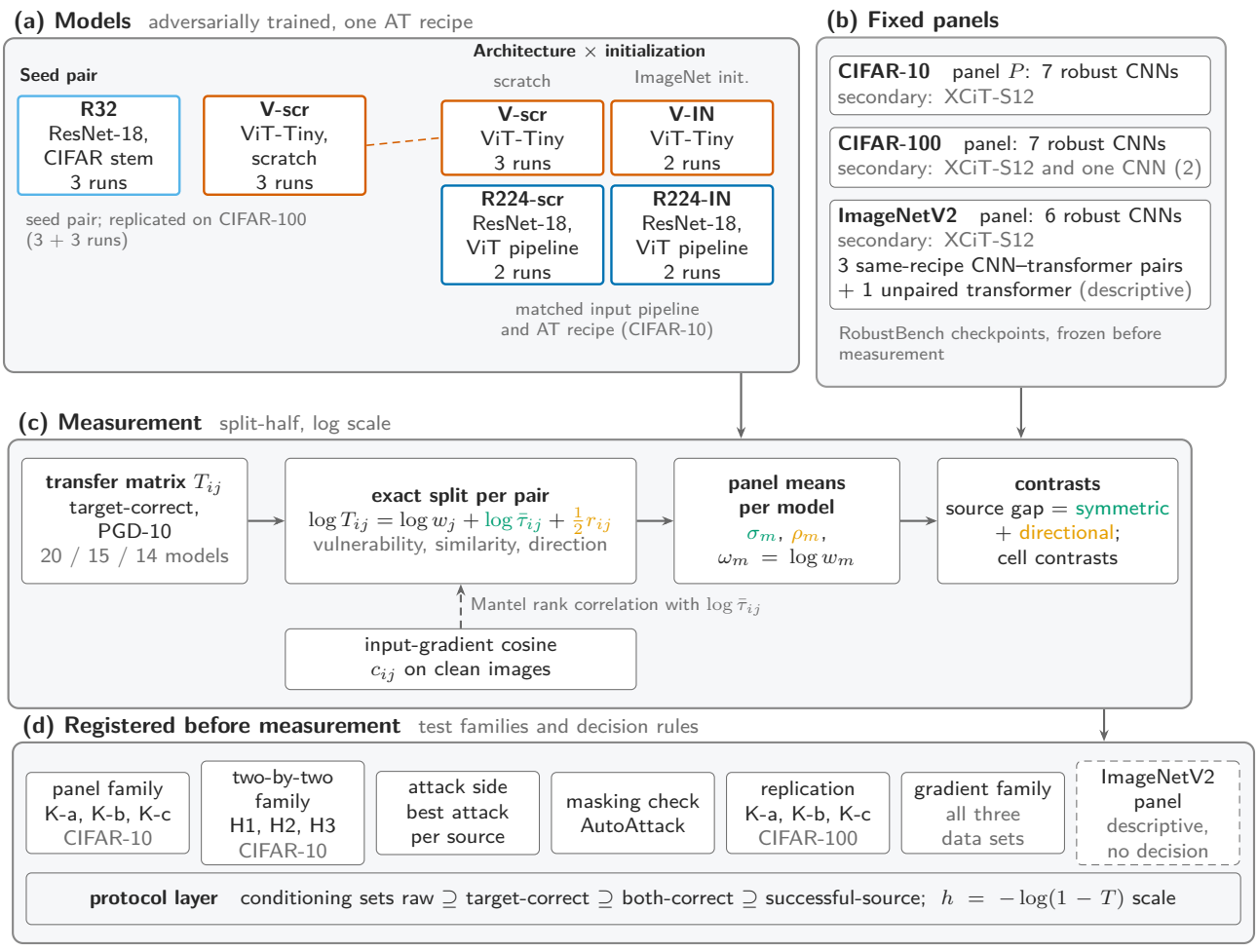

(a) Models adversarially trained, one AT recipe
Architecture × initialization
Seed pair
scratch
ImageNet init.
R32
ResNet-18,
CIFAR stem
3 runs
V-scr
ViT-Tiny,
scratch
3 runs
V-scr
ViT-Tiny
3 runs
V-IN
ViT-Tiny
2 runs
R224-scr
ResNet-18,
ViT pipeline
2 runs
R224-IN
ResNet-18,
ViT pipeline
2 runs
seed pair; replicated on CIFAR-100
(3 + 3 runs)
matched input pipeline
and AT recipe (CIFAR-10)
(b) Fixed panels
CIFAR-10 panel P: 7 robust CNNs
secondary: XCiT-S12
CIFAR-100 panel: 7 robust CNNs
secondary: XCiT-S12 and one CNN (2)
ImageNetV2 panel: 6 robust CNNs
secondary: XCiT-S12
3 same-recipe CNN–transformer pairs
+ 1 unpaired transformer (descriptive)
RobustBench checkpoints, frozen before
measurement
(c) Measurement split-half, log scale
transfer matrix T_{ij}
target-correct,
PGD-10
20 / 15 / 14 models
exact split per pair
$\log T_{ij} = \log w_j + \log \bar{\tau}_{ij} + \frac{1}{2} r_{ij}$
vulnerability, similarity, direction
panel means
per model
σ_m, ρ_m,
$\omega_m = \log w_m$
contrasts
source gap = symmetric
+ directional;
cell contrasts
Mantel rank correlation with $\log \bar{\tau}_{ij}$
input-gradient cosine
c_{ij} on clean images
(d) Registered before measurement test families and decision rules
panel family
K-a, K-b, K-c
CIFAR-10
two-by-two
family
H1, H2, H3
CIFAR-10
attack side
best attack
per source
masking check
AutoAttack
replication
K-a, K-b, K-c
CIFAR-100
gradient family
all three
data sets
ImageNetV2
panel
descriptive,
no decision
protocol layer conditioning sets raw ⊇ target-correct ⊇ both-correct ⊇ successful-source; $h = -\log(1 - T)$ scale